

ZEYING(ING) TIAN

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EDUCATION

Bachelor of Computer Engineering, McGill University
CGPA 3.99 - Dean's Honour List - Full Scholarship

2019 - 2023

PUBLICATION

Kang, Shaofeng, and Zeying Tian. "Optimal Portfolio Construction—A Reinforcement Learning Embedded Bayesian Hierarchical Risk Parity (RL-BHRP) Approach." *arXiv preprint* arXiv:2508.11856 (2025).

Li, Dailun, Zeying Tian, and Yining Duan. "Self-attention on RNN-based text classification." *In International Conference on Computer Network Security and Software Engineering (CNSSE 2022)*, vol. 12290, pp. 20-25. SPIE, 2022.

SKILLS

- Language: C/C++, Golang, Java, Python (Pandas, Numpy, Tensorflow, Scikit-learn, PyTorch), JS/TS
- RPC Frameworks (Thrift, gRPC, kitex), RestFUL Frameworks (Spring, Django)
- Redis, MySQL, Mongo, Vector Databases (Faiss, Viking), Graph Database (ByteGraph)
- Machine Learning Models: XGBoost, VGG, BERT, BART, Seq2Seq, Dual-Tower Models, LSTM, GRU
- Cloud Computing: S3, EC2, ECR, ECS, RDS, AWS Glue, Firehose, Lambda, SNS, QuickSight, Athena

EXPERIENCE

Senior Software Engineer (Level 2-1)

TikTok Inc.

Nov 2023 - Apr 2025

San Jose, CA, U.S.

- Built a similarity-based tracking pipeline for creative asset consumption. Utilized a dual-tower vision model to extract video embeddings and applied ANN (Approximate Nearest Neighbor) algorithms to retrieve similar videos from a vector database within milliseconds. Results were written to ByteGraph for persistent tracking. Enabled quarterly tracking of \$200M in advertising creative spend.
- Developed an ad signature RPC service in C++ (gRPC) for creative deduplication and quality content boosting that could handle 50k QPS. Leveraged a neural network with importance pooling to generate video embeddings, which were used to retrieve similar content from a Faiss database. Content tags were written to a Redis database and served to the ads delivery engine for real-time decision making.
- Integrated ffmpeg-based processing to support Adobe XMP metadata tracking, parsed and stored content metadata in a NoSQL database, and synchronized it via offline pipelines to Hive tables for downstream data engineering tasks. The full pipeline supported tracking of \$60M in creative spend every 2 weeks.

Co-op Software Development Engineer (Intern)

Ericsson

Sept 2022 - Apr 2023

Montreal, Canada

- Worked on the NodeCache for JAMScript for nodes to discover each other in an IoT setting.
- Proved the effectiveness of and implemented milestones.
- Wrote a simulation program in C++ to verify the mobility of nodes in JAMScript.
- Implemented CNN models to detect malicious smart contracts with time-series data.

Software Development Engineer (Intern)

Amazon Inc.

May 2022 - Aug 2022

Toronto, Ontario, Canada

- Designed and implemented a service that archives and creates visual analysis for incoming fulfillment data.

- Wrote the introduction document for the Cratos team that speeds up freshmen onboarding.
- Transformed the incoming SNS data with Lambda and output results to S3 via Firehose.
- Queried results with Athena and presented the results with QuickSight.

Research Assistant (Under Prof. Muthucumaru Maheswaran)

McGill University

Nov 2021 - Jan 2023
Montreal, Quebec, Canada

- Developed the language compiler with the Ohm language parser that translates JAMScript into C and JS.
- Refactored the compiler to increase its flexibility and maintainability.
- Designed CFG grammar and language parser to support ES6 and JAMScript-specific syntax.
- Recorded and generated the incoming program's side effect table and control flow graph.
- Documented the JAMScript's initialization guide, to alleviate the starting process for newcomers.

Software Development Engineer (Intern)

TikTok

Apr 2021 - Aug 2021
Shanghai, China

- Developed a tool to model the landing page's structure in TypeScript, later used in the project refactor.
- Developed a CI which analyzes the reference relationships among project files using Babel.
- Helped develop the interim download page, which generated \$1,000,000 in daily revenue.
- Completed a beginner's guide to a tool, Lynx, that facilitates freshman training in the whole department.

Software Development Engineer (Intern)

Shanghai Shanzhu Technology Ltd.

Apr 2020 - Aug 2020
Shanghai, China

- Developed RESTful backend services with Django and Spring for a hotel management system.
- Developed frontend web pages in Vue.js for a hotel management system.
- Maintained and interacted with MySQL databases which store customers' data.
- Developed .NET applications to read from and write to hotel room cards.
- Wrote unit tests in Java with Cucumber to verify the integrity of hotel backend services.

AWARDS AND CERTIFICATES

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| • Hatch Scholarships (\$10,000), <i>McGill University, Canada</i> | July 2022 |
| • Top 4 teams in Canada, <i>IEEEExtreme 2021, Canada</i> | Nov 2021 |
| • Hatch Scholarships (\$10,000), <i>McGill University, Canada</i> | July 2021 |
| • Excellence Bursaries for ECSE students (\$10,000), <i>McGill University, Canada</i> | Apr 2021 |
| • Top 20 teams in Canada, <i>IEEEExtreme 2020, Canada</i> | Oct 2020 |
| • J B Woodyatt Prize (\$2,000), <i>McGill University, Canada</i> | Aug 2020 |
| • Hatch Scholarships (\$10,000), <i>McGill University, Canada</i> | July 2021 |
| • The Rio Tinto – Richards Evans Intl. Exchange Award (\$10,000), <i>McGill University, Canada</i> | Aug 2020 |